

FIND THE FORMULA

1 Plan A_6

PI set $n = 6$ target;
require exact tests

Student check A_5 ;
test denominators

$$A_5 : 24/24 \text{ exact}$$

no denominator found yet

2 Use a ratio

PI check whether a
denominator is
needed

Student take one frequency
to zero;
use a ratio

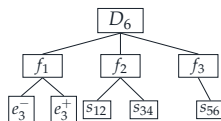
polynomial formula $\rightarrow$ rejected

zero-frequency limit: ratio form

3 Find denominator

PI check factors from
both groups

Student find product inde-
pendently



denominator candidate

4 Simplify formula

PI check powers and
degrees

Student derive D_6 from the
zero-frequency limit

$$D_6 = e_3^- + e_3^+$$

$$\deg N_6 = 11$$

TEST, FIX, AND EXTEND

5 Test numerator

PI check coefficient Q

Student test sums of single-
boundary terms

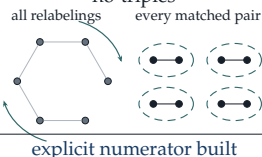
$$\sum \text{isolated walls} \neq Q$$

paired-wall terms required

6 Build N_6

PI require correct sym-
metry
and degree

Student include two-
boundary terms;
no triples



7 Check A_6

PI new BG code:
140/140
in 58 chambers

Student give formula, coeffi-
cients,
and code



NEW A_6 FORMULA ACCEPTED

8 Try $n = 7$

PI check symmetry,
boundaries,
and zero limit

Student find denominator
and
boundary powers

$$n = 6 \rightarrow n \geq 7$$

explicit $N_{n \geq 7}$: not yet found

$$\text{NEW RESULT } A_6 = i 2^5 g^{-3} \frac{N_6}{e_3^- + e_3^+}$$

OPEN

explicit numerator for $n \geq 7$